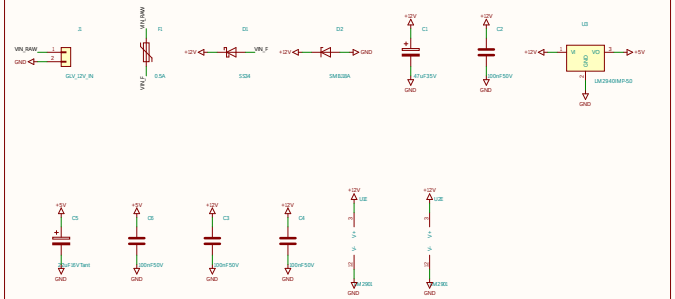
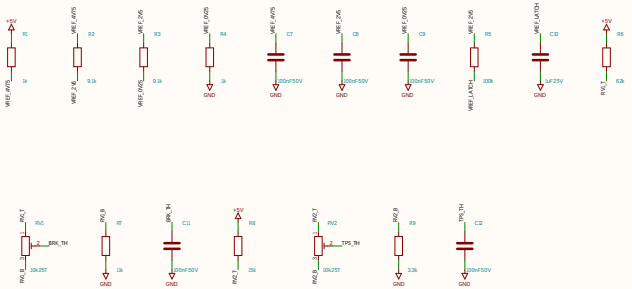
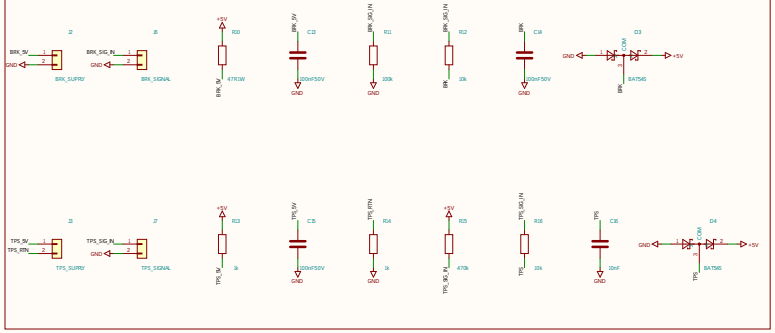
1. GLV POWER INPUT & 5 V REFERENCE SUPPLY



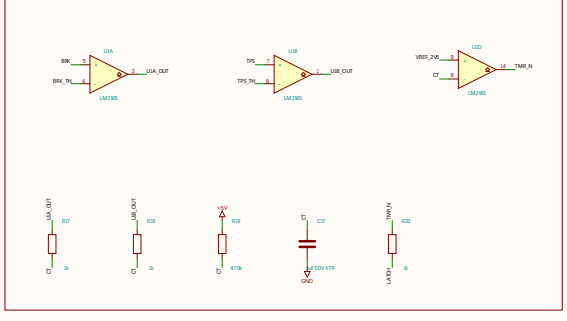
2. REFERENCES & TRIM THRESHOLDS (all ratiometric to +5V)



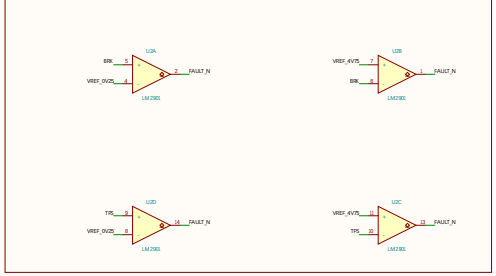
3. SENSOR INPUTS (own sensors; each signal on its own connector, T 11.6.7)



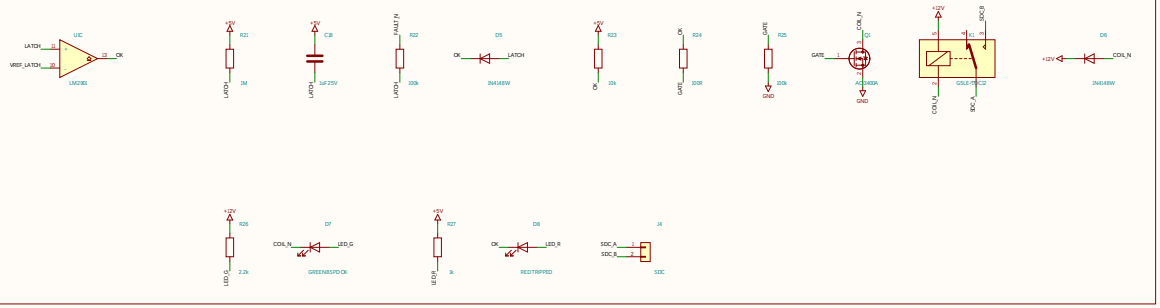
4. PLAUSIBILITY (T 11.6.1) + 500 ms TIMER (T 11.6.2)



5. SCS OPEN/SHORT DETECTION (T 11.9.2)



6. LATCH + SDC RELAY OUTPUT



7. TEST PADS (no connector)



DESIGN NOTES - checked against FS Rules 2025 v1.0 (T 11.6, T 11.9, T 1.3.1, CV 1.4.3, CV 4.1)
Comparators: U1A hard brake | U1B throttle > 20 % | U1D 500 ms timer | U1C latch
U2A/U2B brake signal window | U2D/U2C throttle signal window
T 11.6.1 Trip when BRK > BRK_TH AND TPS > TPS_TH. Latched until LVMS power cycle.
T 11.6.2 Delay 0.693 x R19 x C17 = 326 ms (Monte-Carlo 256-350 ms) < 500 ms.
T 11.6.3 J3 wired DIRECTLY from LVMS output (only fuse/connectors in between).
T 11.6.4 Only interfaces: J1 power, J2/J6 brake sensor, J3/J7 throttle sensor, J4 SDC.
T 11.6.5 RV1 range 0.66-1.27 V = 5.6-26.4 bar (MLH02K 0.5-4.5 V = 0-138 bar); always <= 30 bar.
Set below-wheel-lock pressure. BRK_TH = 0.5 V + 0.029 V/bar x P.
T 11.6.7 Each sensor signal on its own connector (J6, J7) - unplug to test.
T 11.9.2 Open / short-GND / short-5V on any sensor wire -> trip in ~70 ms.
CV 1.4.3 Throttle sensor on the THROTTLE BODY shaft. TPS_TH = V_idle + 0.20 x (V_WOT - V_closed).
CV 4.1.1 K1 COM(3)/NO(3) in series with the SDC; SDC must feed >=2 mechanical relays
(fuel pump relay + ignition/injection relay). CV 4.1.3: de-energised K1 = SDC open.
C5 must be tantalum (LDO stability). Lock RV1/RV2 with paint after setting.

Checked against FS Rules 2025 v1.0: T 11.6, T 11.9, T 1.3.1, CV 1.4.3, CV 4.1
Verified: ngspice behavioural sim + exported-netlist cross-check
CV car, Kawasaki ER-6n 650 (stock ECU). Non-programmable, latching, fail-safe.
ZPR Formula SAE - Electrical

Sheet: /
File: bspd.kicad_sch

Title: Brake System Plausibility Device (BSPD) - Option A

Size: A2 | Date: 2026-11-02 | Rev: B
KiCad E.D.A. kicad (5.1.4)-1 | Id: 1/1

Board Stack Report